CS & IT ENGINEERING



Algorithms

Divide & Conquer



Lecture No.- 03

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Min-Max Algo

Intro to Binary Search.

Topics to be Covered



- Topic
- Topic
- Topic

Binary Search

Merge Sort



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram

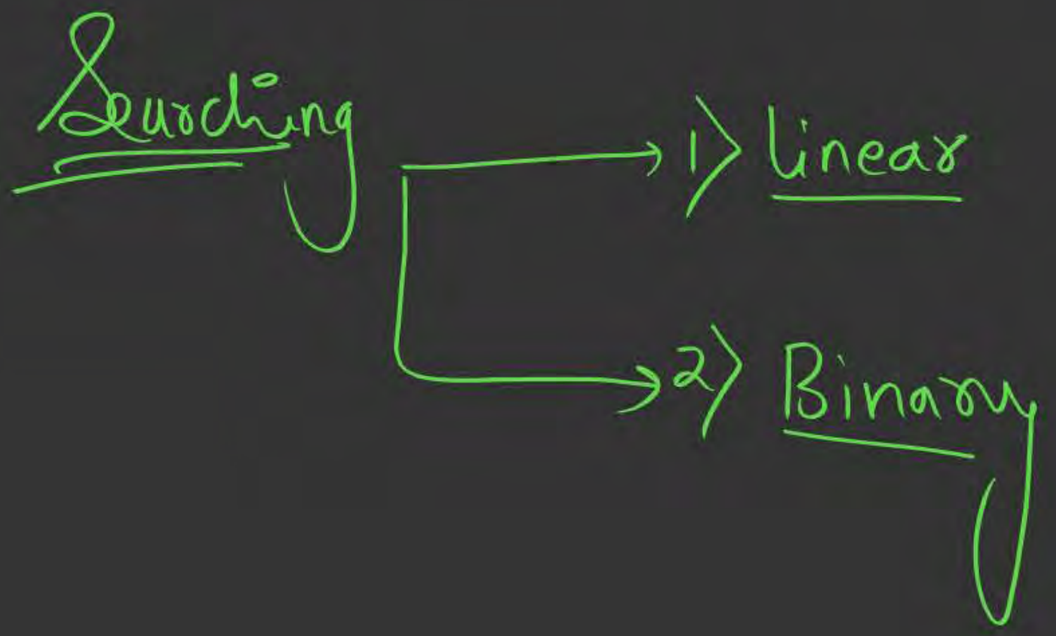


Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

No Algo class

12, 13 May

directly on 14th May.

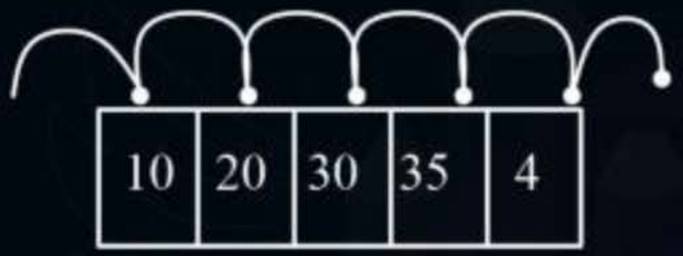

Searching
1) Linear
2) Binary



Topic : Linear Search

Linear Search:-

```
Algo AJ Linear (A, n, x)
For (i  = 1;, i < = n; i++)
if (A[i] = = x)
{
        return i
}
Print (" Not found")
```





Topic : Linear Search

Linear Search TC Analysis:-

(1) Best case: A = | 2 | 20 | 1 | 10 | x = 2

TC = O(1)

(2) Worst case: A = | 2 | 5 | 10 | 7 |

TC = O(n)

Always work

x = 7 or x = 8

elem at last ~~positive~~ index

Not present



Topic : Linear Search

Does linear search take any advantage when the input array is sorted?

→ No

Here are need Binary Search



Topic : Binary Search

Binary search → Divide & Conquer

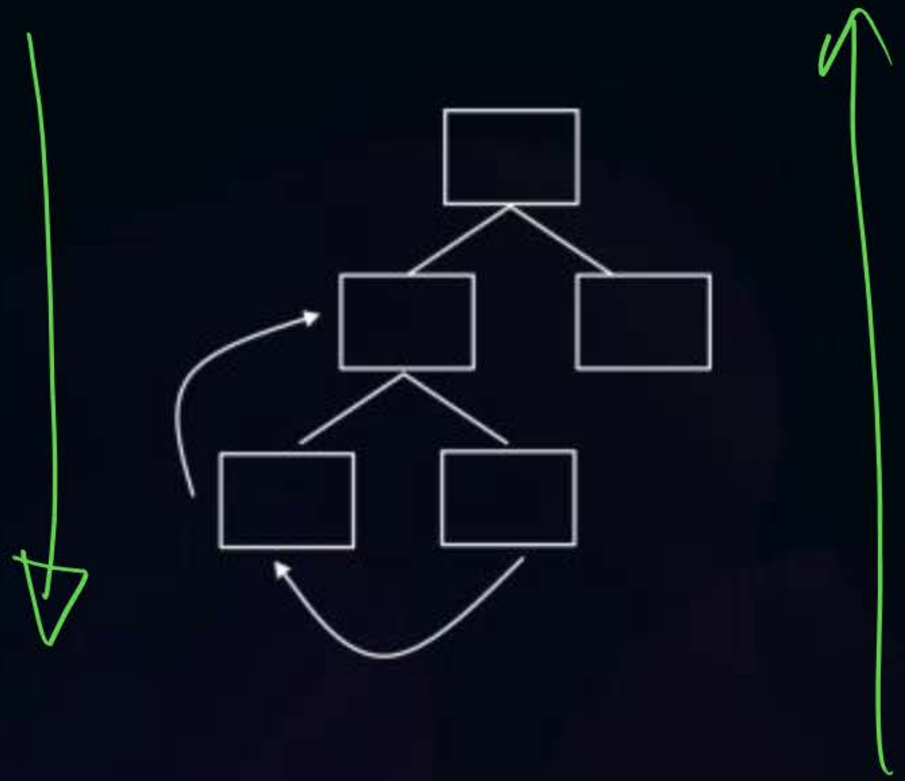
Pre-requisite: Input array should be in sorted order (Mandatory Requirement)

→ No such requirement in linear search.



Topic : Min-Max

Min-max $T(n) = 2T(n/2) + 2$



$T(n) = T(n/2) + T(n/2) + 2$



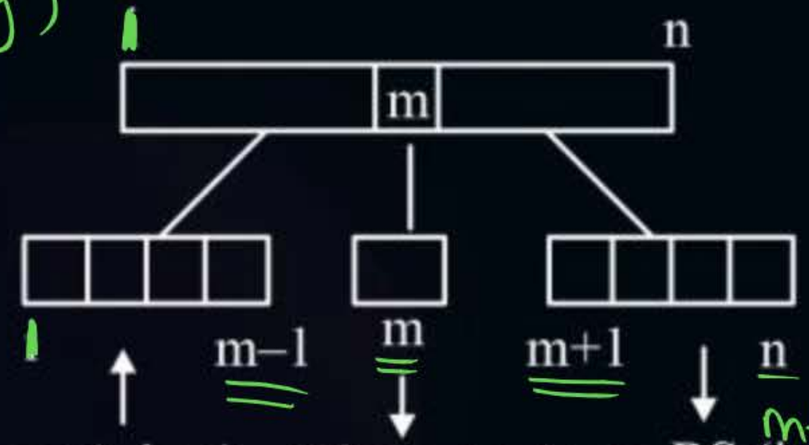
Topic : Min-Max

(6) Algo 30 Jan

GBG 1st Feb

Recursion:

BS(i, n, key)



$x < A[m]$

$m = \left\lfloor \frac{i+n}{2} \right\rfloor$

$(n/2)$

$n/2$

BS (1, m–1, key) BS (m, m, key) BS (~~i~~ m+1, ~~m–1~~ n, key)

Small Prob

OR



a = b

a > b

a < b

a b

2 5

2 < 5

5 5

5 = 5

10 7

10 > 7



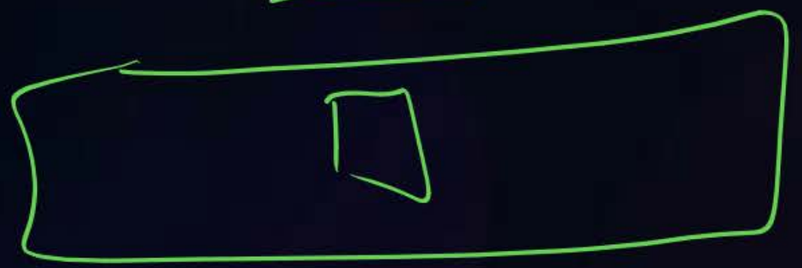
Topic : Min-Max

Recursion:

$$\begin{cases} C1: & Key = A[m] & \rightarrow \text{Small prob sorted} \\ C2: & Key > A[m] & \rightarrow \text{Explore BS(m} + 1, \text{n, key)} \\ C3 & key < A[m] & \rightarrow \text{Explore BS(1, m} - 1, \text{key)} \end{cases}$$

Soln (Stop)

$x > A[m]$



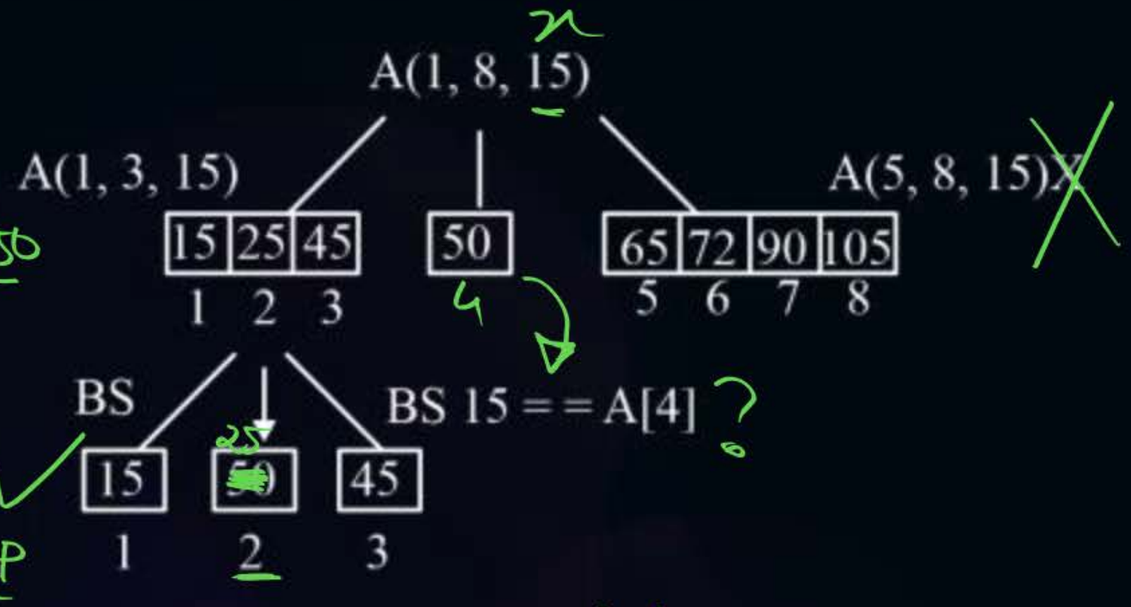
Topic : Min-Max

Eg.

15	25	45	50	65	72	90	105
1	2	3	4	5	6	7	8

$x = 15$

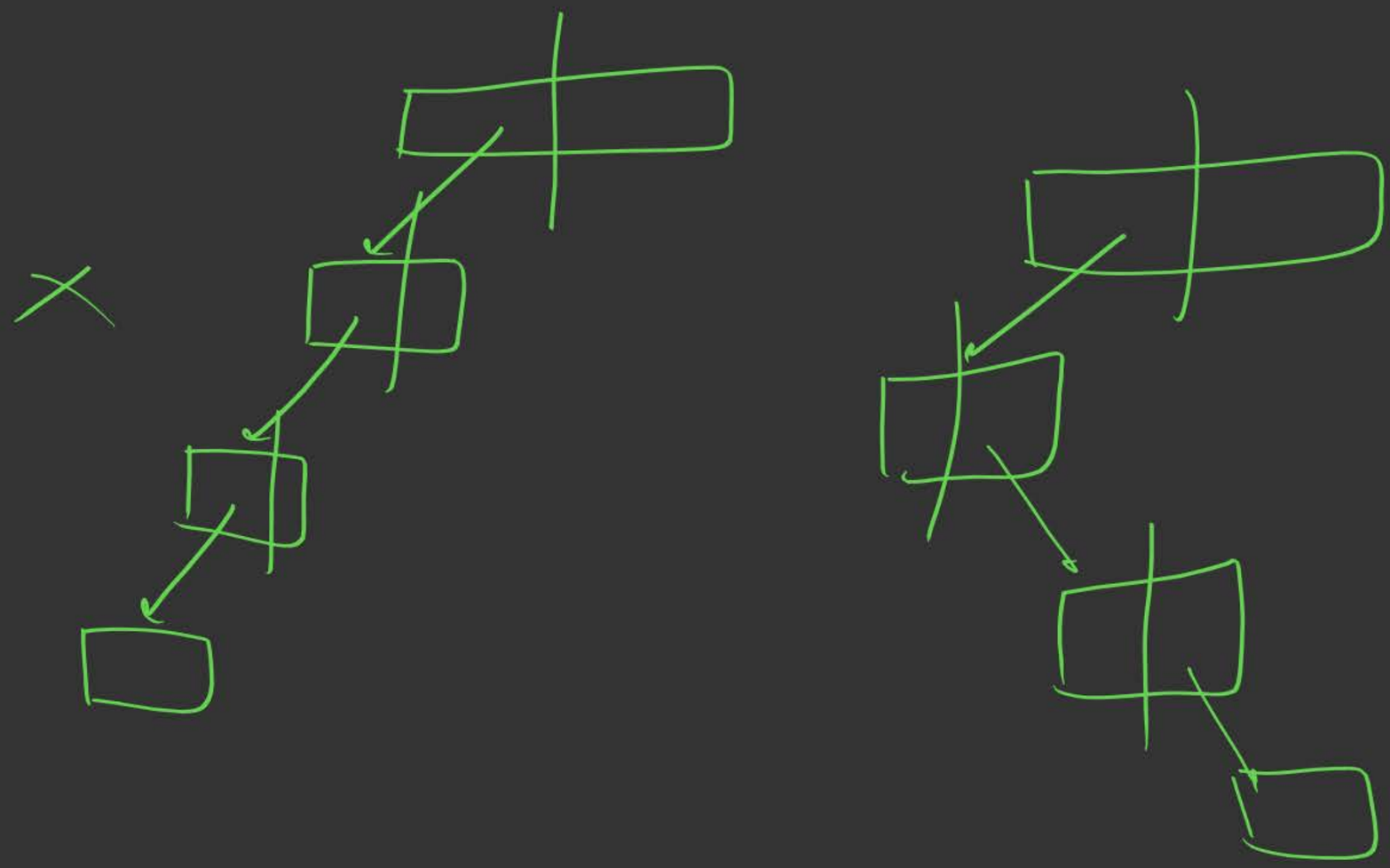
Idea:

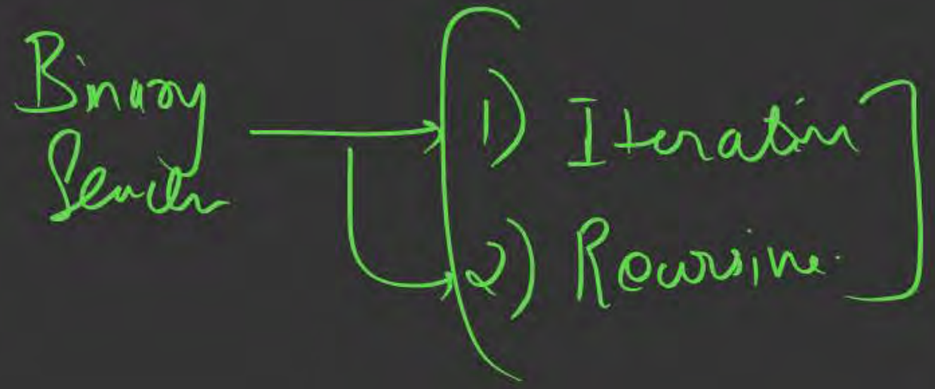


$A(1, 8, 15)$

$\left\lfloor \frac{1+8}{2} \right\rfloor = \left\lfloor \frac{9}{2} \right\rfloor = 4$

{We will either ~~recursively~~ recursively explore left subproblem or right subproblem BUT ~~here~~ never both}



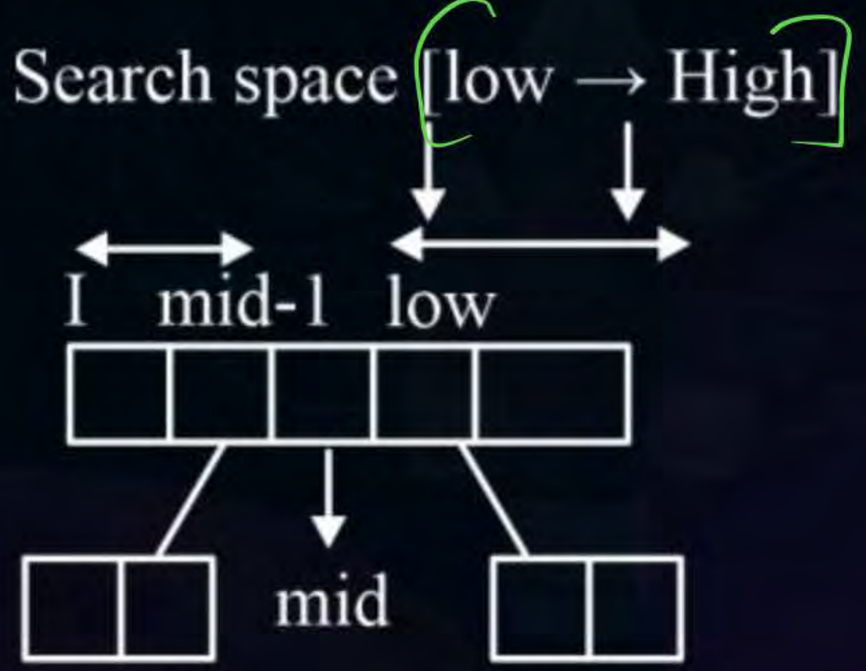

Binary
Search
1) Iterative
2) Recursive



Topic : Divide and Conquer

Non-recursive/Iterative

```
Algo Binary search (A, n, x)
// A[1..n] → non-decreasing order
// return index of x, else o.
Low = 1, high = n
While (low ≤ high)
        mid = ⌊(low+high)/2⌋
if(A[mid] > x)
        high = mid-1
else if (A[mid] < (x)
        low = mid+1
else
        Return mid
```





Topic : Divide and Conquer

Recursive

```
1.  Algorithm BinSrch(a, i, l, x)
2.  // Given an array a[i: l] of elements in non decreasing
3.  // order, 1 ≤ i ≤ l, determine whether x is present, and
4.   // if so, return j such that x = a[j]; else return 0.
5.  {
6.  if (l = i) then // If Small(P)
7.  {
8.  if (x = a[i]) then return i;
9.  else return 0;
10. }
```



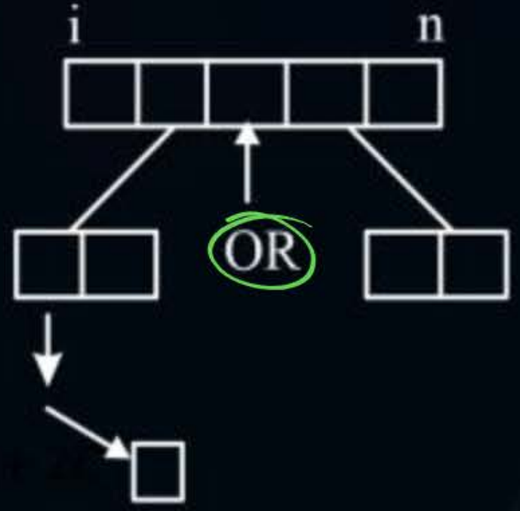


Topic : Divide and Conquer

```
11.  else
12.  { // Reduce P into a smaller subproblem.
13.  mid := [(i + l)/2];
14.   if (x = a[mid]) then return mid;
15.  else if (x < a[mid]) then
16.          return BinSrch(a, i, mid - 1, x);
17.  else return BinSrch(a,mid + 1, l, x);
18.  }
19.  }
```



Topic : Divide and Conquer



Logic:

$T(n) = T(n/2) + C, n > 1$

$= b, n = 1$

$T(n/2) = T(n/2^2) + C$

$T(n) = T(n/2^2) + 2C$

$= T(n/2^3) + 3C$

General:

$T(n) = (n/2^k) + k*c$

for B.C, $\frac{n}{2^k} = 1 \Rightarrow 2^k = n$

$K = \log_2 n$

$T(n) = T(1) + k * c$

$= b + c * \log_2 n$

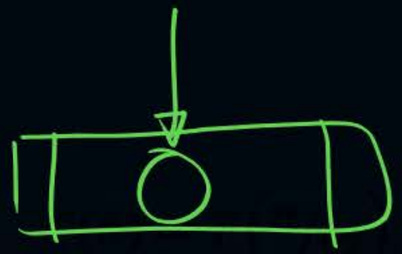
$T(n) = O(\log_2 n)$



Topic : Divide and Conquer

Hence, In Binary search there is only divide but not combine no need

TC analysis :-



B.C = $\Omega(1)$

W.C = $O(\log n)$

i/p: Sorted

	B.C	W.C
1) L.S	$\Omega(1)$	$O(n)$
2) B.S	$\Omega(1)$	$O(\log n)$



Topic : Divide and Conquer

Space Complexity analysis

(1) Non-Recursive

$\Rightarrow$ SC = O(1)

(2) Recursive

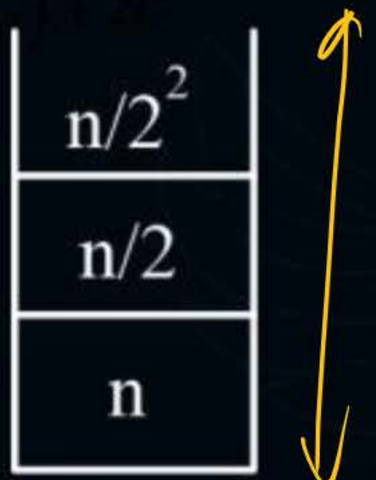
SC = $O(\log_2 n)$

↓

Recursion stacks

$k = \log_2 n$

$n/2^k = 1$



③ Merge Sort



Topic : Divide and Conquer

Merge Sort

(3) Merge sort

Divide & Conquer based sorting Technique

Conquer → Merging Aglo (Principle of Merging)





Topic : Divide and Conquer

Principle of Merging (Conquer)

L1 → m
L2 → n

Problem Statement:

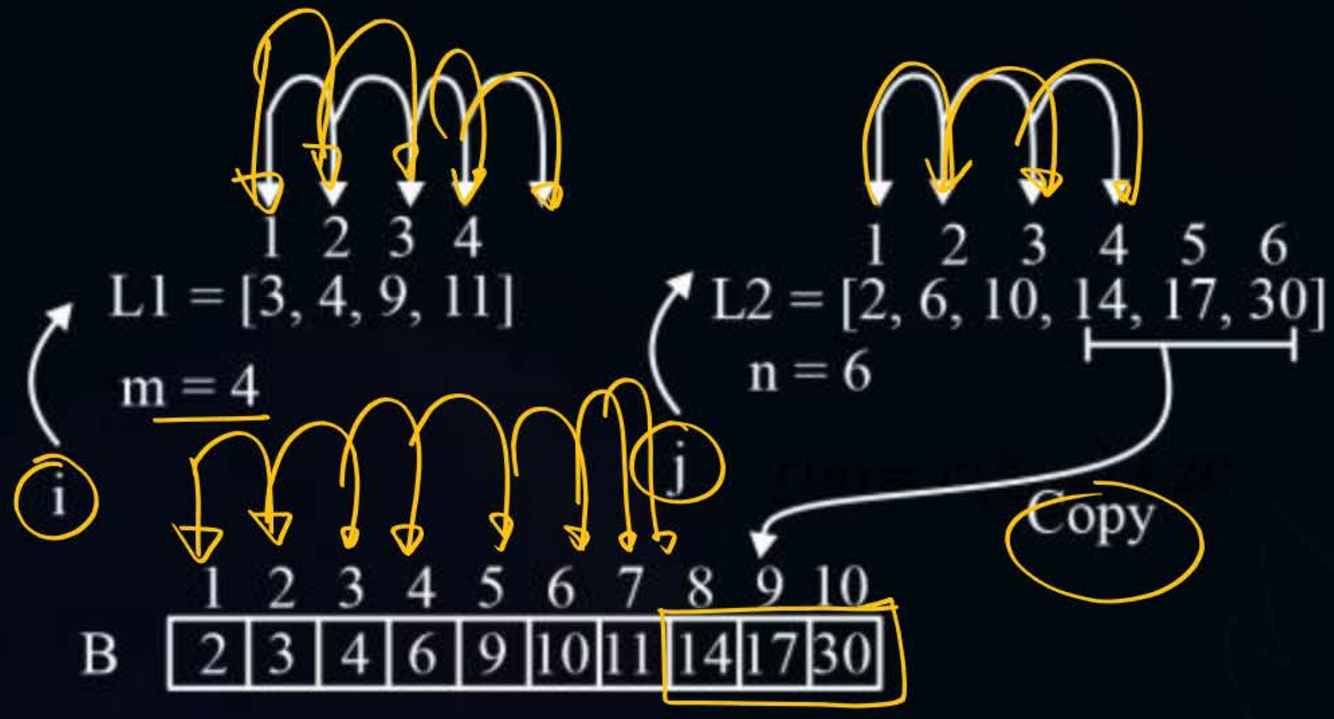
Given two sorted lists/Arrays L1 & L2 such that m ≤ n, Merge them into a single sorted list/Array → Size ⇒ (m + n)

(2-way merging)



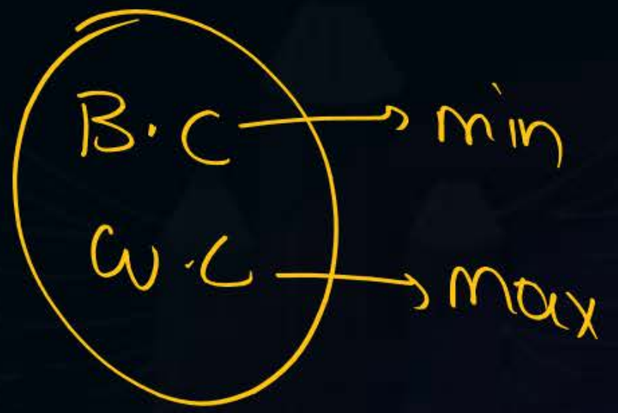
Topic : Divide and Conquer

Eg.



(m+n)

30%





Topic : Min-Max Problem

#Q. What are the min & max number of comparisons ~~read~~ needed in 2-way merging?

Sorted L1 → m L2 → n & m ≤ n

70 %

default

A. m, m + n

B. n, m+n–1

C. m, m + n–1

D. n, m + n



Topic : Divide and Conquer

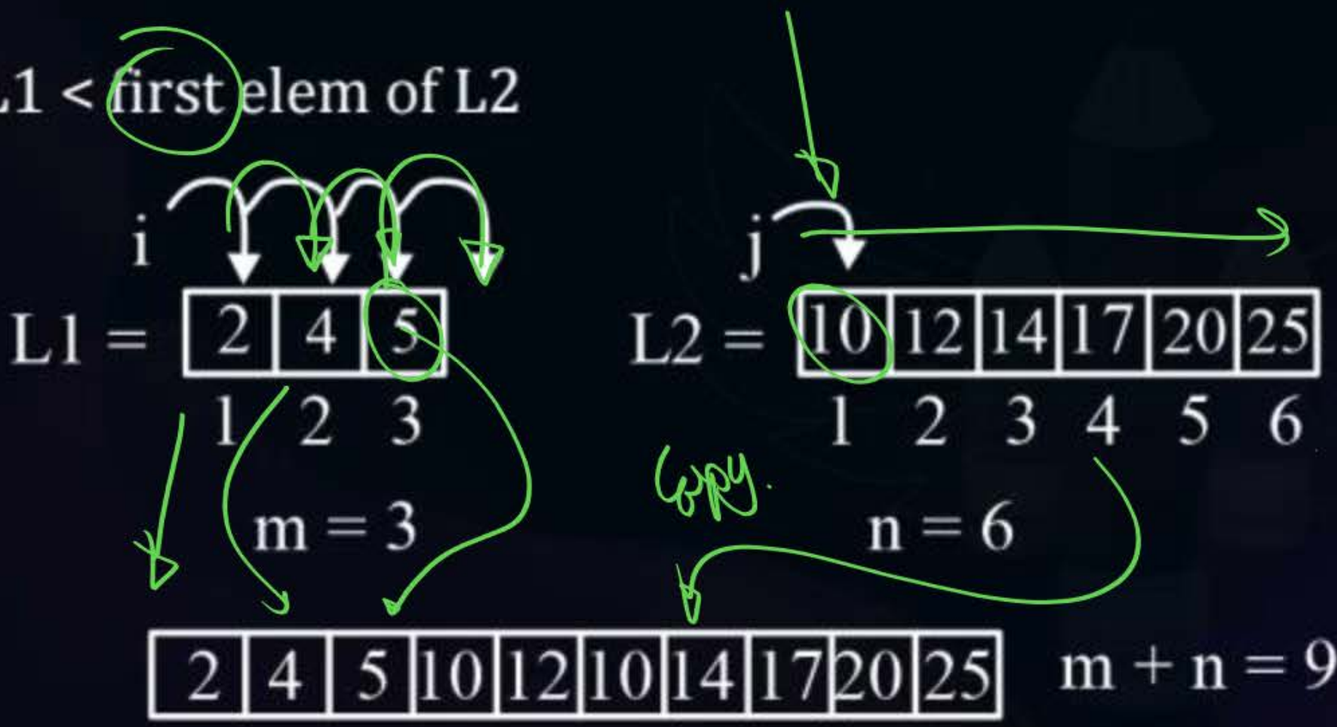
(1) Min comparisons::

Input: When all elems of smaller (m) list (L1) are lessen than all the elems of larger (n) list (L2)

OR

When last elem of L1 < first elem of L2

min (m, n)





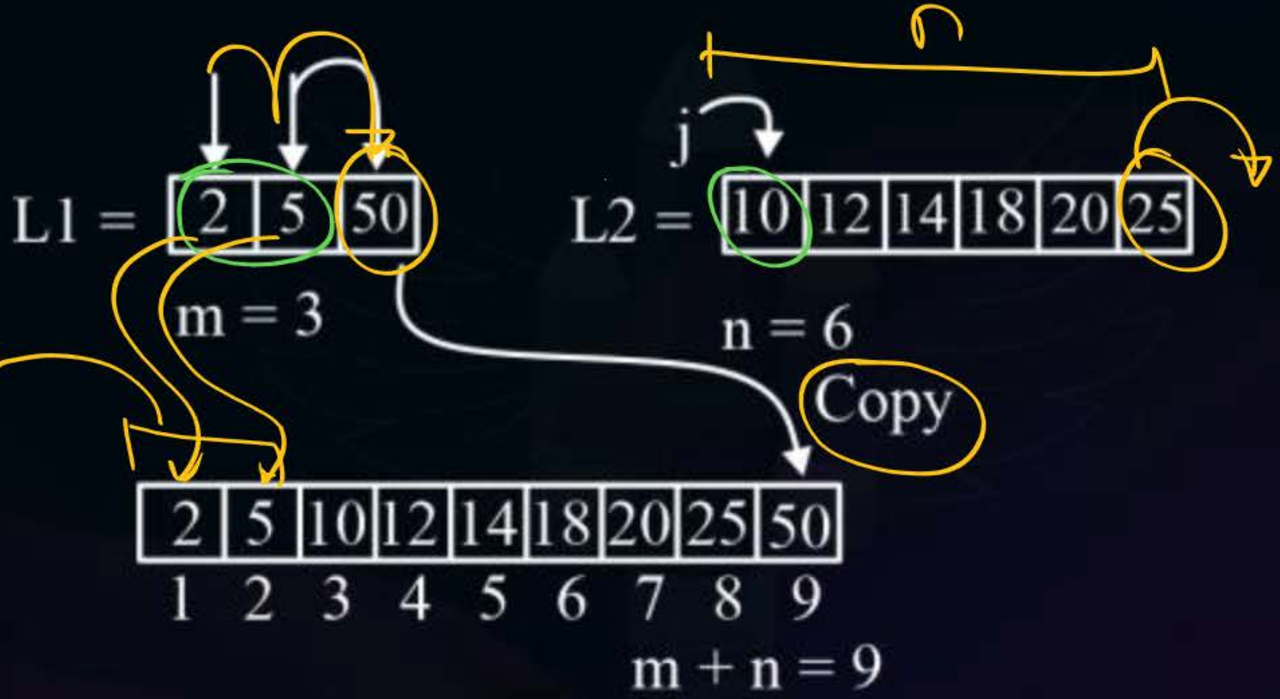
Topic : Divide and Conquer

(2) Max comparisons::

Input: Smaller List (L1): (1 → m – 1) < first elem of larger list (L2)

and m^{th} elem of L1 (last elem) > all elems of L2 (last elem of L2)

Total elem comparisons
= (m – 1) + n
= (m + n – 1)
max comparisons





Topic : Divide and Conquer

Summary:-

The number of elem comparisons reqd to merge two sorted list into a single sorted list ranger between

$$m \leq comp \leq (m + n - 1)$$

Sorted

$L1 \rightarrow m$

$L2 \rightarrow n$

~~$L3 \leq n$~~

$m \leq n$

general.

$$\min(m,n) \leq Comp \leq (m+n-1)$$



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Searching → L.S
→ B.S

Merging Algo.



THANK - YOU